

CO2 - Assessment Tool-2

① A) Given dataset,

i) identify missing values.

Cust ID	Age	Monthly Spend	Purchase Freq.
C2	missing	35	8
C3	31	missing	10
C5	27	25	missing

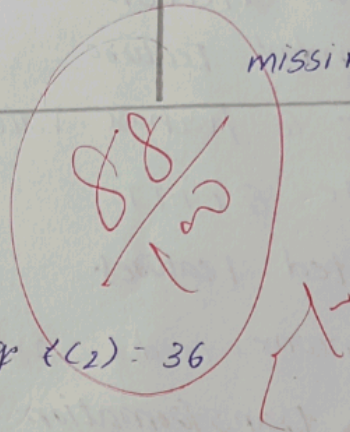
ii) Apply simulation techniques.

Median: 23, 31, 105, 27, 40, 36, 42

sorted: 23, 27, 31, 40, 42, 105

Median age = (36)

Missing age (C2) = 36



B) Outlier Detection and treatment

i) z-score method

check if $|z| > 3$ ($z = \frac{x - \mu}{\sigma}$)

if $> +4n$;

ii) outlier records

C4: Age = 105, Spend (500)

C8: Spend = (300)

C) Final cleaned data.

C-ID	Age	Gender	Monthly Spend	Purchase
C1	24	40	20	5
C2	34	60	35	8
C3	44	85	38	10
C4	52	120	100	50
C5	29	45	2.5	10
C6	36	150	100	12

② Given dataset,

A) i) covariance

mean income = 80625

mean purchase = 198.75

covariance = 3528.22

iii) importance

→ covariance helps identify relationships b/w features and detect variables.

ii) Interception

→ covariance is positive,
so increases

B) Feature Selection

i) Redundant Feature

→ income & purchase - Value

→ income & EMJ

Justification

→ Removes redundant features, reduces

ii) Selected Features.

Age, Income, credit, orders

c) Data transformation

i) Standardization. use formula.

$$Z = \left(\frac{x - \mu}{\sigma} \right)$$

ii) why standardization

→ Brings all features to some scale

→ improves model Accuracy

③ A) Equal width Binning

→ Study - Hours

Min = 2, max = 11

Bin width = $(11 - 2) / 3 = (3)$

Bins

Bin 1: (2 - 5)

Bin 2: (5 - 8)

Bin 3: (8 - 11)

②

Bin1: 2, 3, 4, 5

Bin2: 6, 7, 8

Bin3: 9, 10, 11.

B) Equal-Frequency Binning

Bin1: 2, 3, 4, 5

Bin2: 6, 7, 8

Total Score

Bin1: 60, 50, 55, 65

Bin2: 75, 70, 90

Bin3: 95, 98, 100

C) concept hierarchy

Study hours

low: (2-4) , Medium: (5-11)

High: (8-11)

Test Score

poor - (40-60) Average = (61-80)

Excellent: (81-100)

④ Data Transformation in online Shopping Behaviour Analysis.

A) Equal Frequency Binning

Divide into 3 bins

Bin1: (15, 30, 45, 60)

Bin2: (120, 300, 600)

Analysis

- Groups customers with similar visit duration
- Separates short, medium and long

B) Z-score Normalization

i) Mean and SD

Mean: $(\mu) = (74.7 \text{ seconds})$

SD: $(\sigma) = (1112.8 \text{ second})$

Formula:

$$Z = \left(\frac{x - \mu}{\sigma} \right)$$

$$3600 \text{ Sec} \rightarrow (Z) = (2.56)$$

ii) comparision

→ Z-score → best for handling outliers and data with different States.

